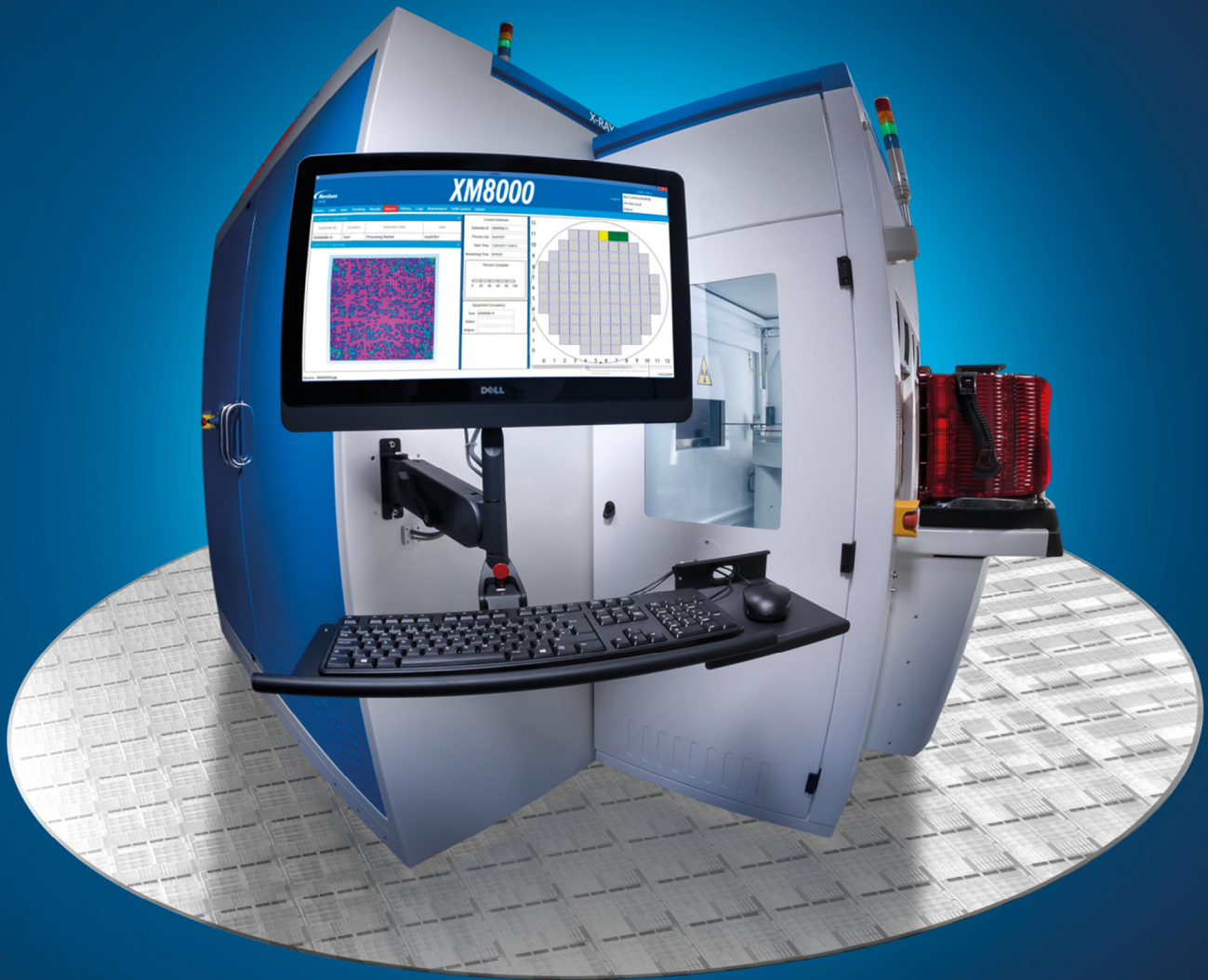


Complexity Simplified



XM8000

Intelligent X-ray Metrology

User Guide

www.nordson.com/TestInspect


Nordson
TEST & INSPECTION

Important information for our customers

Dear Customer,

Congratulations on your purchase of your Nordson Test and Inspection test equipment.

Nordson Test and Inspection is committed to excellent service worldwide and our goal is to provide you with professional assistance in the use of our Test and Measurement Equipment, wherever you are located.

Nordson Test and Inspection offices are located in strategic centers in the USA, Asia and Europe and the company's customer service and support is acknowledged to be 'the best in the business', with global cover on a local level in most countries.

We trust that you find our product an invaluable tool and hope that you will pass on any comments you may have so that we can continue to enhance the product and service to you.

Our web address is: www.nordsondage.com.

Foreword

The products and equipment described in this guide are subject to continuous development and improvement. Consequently, there may be minor variations in specifications, facilities or operation that are not covered in this guide.

The installation, operation and maintenance of this equipment must only be performed by suitably trained and qualified personnel.

Every effort has been made to ensure that the information provided in this guide is accurate at the time of going to print. If any errors or omissions are noticed, please notify Dage Precision Industries Ltd.

Copyright

© 2022 Dage Precision Industries Ltd. All rights reserved.

No part of this publication may be reproduced, transmitted, stored in a retrieval system, or translated into any language in any form by any means without the prior written permission of Dage Precision Industries Ltd.

Trademarks

All trademarks are acknowledged as belonging to their respective companies.

Patents

Please note that Nordson Test and Inspection products are patent protected and covered by the patents listed at <https://www.nordson.com>.

Contact information

Dage Precision Industries Limited,
25 Faraday Road,
Rabans Lane Industrial Area,
Aylesbury,
Buckinghamshire,
HP19 8RY,
United Kingdom.

Regional offices

	Sales	Customer Service
UK & Europe	globalsales@nordsondage.com +44 1296 317800	DPI-customerservicex@nordsondage.com
Germany	sales.de@nordsondage.com +49 89 2000 338 270	ServiceDageGermany@nordsondage.com
South East Asia	sales.sg@nordsondage.com +65 6552 7533	dageseas-ray@nordson.com
Japan	sales.jp@nordsondage.com +81 120 537 555	japan.at.cs@nordson.com
China	sales.ch@nordsondage.com +86 512 6665 2008	chinaservicex@nordsondage.com
Americas	sales@nordsondage.com +1 760 930 3307	serviceUSA@nordsondage.com
Taiwan	globalsales@nordsondage.com +886 2 2902 1860	Contact Sales
Korea	korea.at.cs@nordson.com +82 31 462 9642	Contact Sales

Software license

Dage Precision Industries Ltd Software License

1. You, the end user, are granted a non-exclusive and non-transferable license to use the software accompanying Nordson Test and Inspection products only in connection with those products. No license is granted for any other purpose. There is no right granted to copy, redistribute, reverse engineer, or modify the software.
2. The software, including any source code, documentation, appearance, structure and organisation, are proprietary products of Nordson Test and Inspection and are protected by copyright and other laws.
3. This license is not a sale. Title and copyrights to the software, including source code, documentation, appearance, structure, organisation and any copies made by you remain with Nordson Test and Inspection.
4. This Agreement is effective until terminated. Failure to comply with the terms and conditions of this Agreement will result in the automatic termination of this Agreement and will make available to Nordson Test and Inspection any and all legal remedies. Upon termination of this license you must immediately destroy the software programs and all back-up copies thereof.

About this guide

This guide describes the operation of the Nordson Test and Inspection XM8000 X-ray machine. It covers the three alternate-specification models:

- XM8000 3D IC
- XM8000-5 WB
- XM8000-7 TSV

It is intended for use by suitably-qualified and trained operators.

For preinstallation and maintenance instruction, refer to the Preinstallation and Maintenance Guides.

Document revision: 2 EN, January 2022

Document number: DAGE-0061

The English version of this document is considered to be the original instructions.

The following documents are associated with this guide:

- XM8000 Preinstallation Guide
 - XM8000 Maintenance Guide
 - Nordson Test and Inspection installation overview drawing XM8000 - 803824
 - Brooks JCP EFEM User Guide
-
- Third-party documents for:
 - PC system
 - Monitor

Design variations

Due to the Nordson Test and Inspection policy of continuous product development and improvement, some of the illustrations provided in this document may differ from the actual system you are working on. However, most of these variations are cosmetic and do not impact on the associated procedure steps. Any major differences will be mentioned, where applicable.

Radiation units

The unit of radiation measurement used throughout this guide is defined by the International System of units (SI): Sievert (Sv). However, some countries (for example, the USA), use alternative units. The conversion to REM is:

1 μ Sv per hour is equivalent to 0.1 mrem per hour.

Contents

1.	Safety information	3
1.1.	Document signal words	3
1.2.	Labels	4
1.3.	Operators safety rules	7
1.4.	Hazardous Energies	7
1.5.	Safety interlocks	8
1.6.	X-ray safety	9
1.7.	Statutory regulations	10
1.8.	Local safety rules	10
1.9.	Routine checks of X-ray cabinet	10
2.	Machine overview	11
2.1.	Machine layout	12
2.2.	Interior layout	15
2.3.	Manipulator layout	16
2.4.	Software interface	18
2.5.	Machine components	21
3.	System operations	23
3.1.	Manually loading a carrier	24
3.2.	Logging in	24
3.3.	Using the Home tab	25
3.4.	Viewing a load port state	26
3.5.	Viewing jobs	28
3.6.	Manually starting a process	28
3.7.	Managing an ad hoc process	29
3.8.	Tracking a substrate	31
3.9.	Viewing job results	31
3.10.	Transferring job results	35
3.11.	Viewing system alarms	36
3.12.	Viewing job history	36
3.13.	Configuring local or facility host control	37
3.14.	Configuring communications	37
3.15.	Managing host and user messages	38
3.16.	Changing a password	38
3.17.	Managing users	39
3.18.	Specifying log capture	39

4.	System maintenance operations	41
4.1.	Maintenance control tab	41
4.1.1.	Equipment status	42
4.1.2.	Manual control	42
4.1.3.	Recovery	44
4.1.4.	Door control	45
4.1.5.	IO Control	45
4.1.6.	LOAD port usage	46
4.1.7.	Equipment constants	47
5.	Troubleshooting	49
6.	Index	51

1. Safety information

1.1. Document signal words

The following Signal Words are used in this document. All safety related notes and reminders are marked by an appropriate symbol and the text highlighted in RED.

Please read them carefully:

Symbol	Meaning
 DANGER	DANGER indicates an imminently hazardous situation which, if not avoided, will result in death or serious injury.
 WARNING	WARNING indicates a potentially hazardous situation which, if not avoided, could result in death or serious injury.
 CAUTION	CAUTION indicates a potentially hazardous situation which, if not avoided, may result in minor or moderate injury. It may also be used to alert against unsafe practices.

1.2. Labels

Safety labels are attached to the machine to indicate potential hazards.

It is **very important** you take note of these safety labels, and follow the specified safety procedures and instructions printed on them.

In certain circumstances a further label may be attached underneath, or adjacent to each safety label. This is to provide a local language translation of the safety label text.

Label	Meaning
	For the purposes of this unit, this indicates that X-rays are emitted when the X-ray tube is energized. X-rays are harmful to the human body. To comply with local and national regulations additional text may be required to accompany the radiation trefoil. Due to variations in local requirements this will need to be supplied locally.
	This indicates the presence of high voltages within the equipment. High voltages can cause death or serious injury.
	This indicates the presence of poisonous material within the machine.
Label	Meaning
 ⚠ WARNING Ionising Radiation Exposure to ionising radiation when energised may cause death or serious injury. Lockout / tagout before servicing.	This indicates that X-rays are emitted when the X-ray tube is energized. X-rays are harmful to the human body.
 ⚠ WARNING Toxic Lead (Pb) Lead is a poisonous metal that is harmful to the human body. Avoid direct contact with the skin.	This indicates the presence of a poisonous material within the machine. Lead is a poisonous metal that is harmful to the human body. Avoid direct contact with the skin.

Label	Meaning
 ⚠ WARNING Hazardous Voltage Enclosed Contact may cause shock, burns or death. Disconnect and lockout power before servicing.	This indicates the presence of high voltages within the equipment. High voltages can cause death or serious injury.
 ⚠ WARNING Hazardous Voltage Inside. Disconnect and lockout power before opening.	This indicates the presence of high voltages behind doors. High voltages can cause death or serious injury.
 ⚠ WARNING Live after EMO. Risk of electric shock or burn. Turn off and lockout main disconnect before servicing.	This indicates the presence of high voltages within the equipment after Emergency shut down. High voltages can cause death or serious injury.
 ⚠ WARNING Hazardous Voltage Present with Cable Disconnected. After machine power-off, residual high voltage may be present. Voltage or current sufficient to cause shock, burn or death. Before disconnecting, power down machine, lockout / tagout, then wait 5 minutes. When handling, always: Discharge metal strip to earth. Avoid contact with metal tip.	This indicates the presence of residual high voltage after power off and electrical lockout. High voltages can cause death or serious injury.
 ⚠ WARNING Moving Parts Present Can result in serious injury to hands and fingers. Keep hands away from moving parts. Lockout / tagout before servicing.	This indicates that there are moving parts / robots in this area. There is a possibility that hands or fingers may get caught, resulting in injury or crushing.
 ⚠ WARNING Mechanical Hazard Moving Parts and pinch points. Keep hands clear during operation. Lockout / tagout before servicing.	This indicates mechanical automation moving hazard. There is a possibility that hands or fingers may get caught, resulting in injury or crushing.
 ⚠ WARNING Crush Hazard Heavy object. Before withdrawing, securely fit floor support tracks. May cause death or serious injury. Approx. Weight - 750kg	This indicates that the object weighs approximately 750kg. Ensure the support tracks are securely fitted before moving.

Label	Meaning
 WARNING Heavy Object (25kg) Risk of injury or crushing. Observe all lifting precautions.	This indicates that the item weighs 25kg. Mishandling during lifting could result in injury or crushing.
 CAUTION Heavy Object Can cause muscle strain or back injury. Use lifting aids and proper lifting techniques.	This indicates that the item weighs in excess of 25kg. Mishandling during lifting could result in muscle strain or back injury.
 CAUTION Hot Surface To avoid possible skin burns allow surface to cool before touching. Lockout / tagout before servicing.	This indicates that during operation the surface may become very hot. Even after the equipment has stopped operating, some components may remain hot for some time. Confirm that they are cool before touching.

1.3. Operators safety rules

To avoid damage to the machine as well as avoid the risk of personal injury:

- Always** read this guide to become fully acquainted with all aspects of using the machine
- Always** observe all relevant local regulations and follow normal operational good practice for handling compressed air
- Always** isolate the machine from electrical and pneumatic supplies using the Lock out - Tag out procedures before carrying out any maintenance work
- Always** maintain the machine as prescribed in the Maintenance Guide
- Always** adhere to Local Rules when operating the machine to ensure there is no risk due to radiation leakage
- Always** follow the recommended cleaning procedures and clean the machine on a regular basis

- Do not** place magnetic samples inside the machine as this will cause tube damage and invalidate the warranty.
- Do not** attempt to operate the machine if you are unqualified, or unfit to do so
- Do not** defeat any of the machine's safety features
- Do not** use the machine if it is damaged or faulty in any way, or where there is any doubt about its safe operation
- Do not** use the machine for any purpose other than as specified in this guide
- Do not** attempt to operate the machine with the doors open by overriding the interlocks
- Do not** compromise the integrity of the cabinet's lead shielding
- Do not** make any modifications to the machine without the express permission of Nordson Test and Inspection.

1.4. Hazardous Energies

Once the machine has been connected to the relevant service, the following hazardous energies are present within the equipment:

- Electricity
- Compressed Air
- Vacuum

And when energised:

- X-ray ionising radiation

1.5. Safety interlocks

The machine can be powered up only when:

- All Emergency Stop (EMO) pushbutton switches are in the released position.
- The machine's main electrical power isolator is in the on position.
- The key is inserted into the key-operated switch and turned to either of the 'on' positions:
 - Power Enable - the control computers, pneumatics and motors are enabled, but all X-ray systems are disabled.
 - X-Ray Enable - all machine systems are enabled, including X-ray generation. However, X-rays can only be generated when the machine is powered up, the key switch is turned to X-Ray Enable, and the interlocks on both the maintenance/service access doors and the EFEM to X-ray cabinet shutter are closed.



Note: The door and shutter interlocks are double-redundant, self-monitoring systems designed for safety critical applications.

The maintenance/service access doors are locked by a latch that can only be released by the user either via the software interface, or by operating the manual override actuator located in the electrical compartment. Either way, the lock latch releases the doors which activates the interlocks. This system:

- prevents uncontrolled opening of the doors with the X-rays on, which is extremely hazardous to health.
- turns the X-rays off.
- allows the system to bring all motion to a controlled stop.

1.6. X-ray safety

Ionizing radiation



The machine produces X-rays which are harmful to the human body.

The X-ray source is completely contained in the shielded cabinet, with interlocking access doors preventing accidental exposure in normal use. When the interlocks are activated, or during an electrical lockout, there is no possibility of any residual ionizing radiation remaining inside the shielded cabinet. X-rays cease to be produced as soon as the power is removed from the X-ray tube.

Illuminated X-ray on warning indicators are provided on the front, side and rear panels of the cabinet.

An underlying principle of radiation protection is that the radiation from the machine should be limited to 'As Low As Reasonably Achievable'. This principle is embodied in the machine design, but this is also a requirement of the machine users, and appropriate guidance should be given in the 'safe working arrangements' or 'local rules'.



The machine must be checked at regular intervals to verify that X-ray leakage from the shielded cabinet during operation is no higher than 1 μ Sv/H. Refer to the Maintenance Guide for details.

If radiation is detected above this level or a fault suspected, **switch off the machine immediately.**

Shielding methods

The machine is lined with lead in all areas where X-rays could otherwise escape through the cabinet. The thickness of the lead-lining varies according to the power of the expected radiation.

The cabinet maintenance/service access doors and the EFEM to X-ray cabinet shutter are also lead-lined and overlap their respective aperture in the cabinet to ensure complete prevention of X-ray leakage. Shield bars are used extensively around the openings to reduce/eliminate radiation leakage.

X-ray On indicators

The machine is fitted with a total of five pairs of long-life LED warning indicators that are illuminated when X-rays are being generated within the cabinet. Four pairs are located around the four upper panels of the X-ray cabinet and one pair is located at the front of the EFEM unit. These indicators provide easy visibility of X-ray generation from all four sides.

1.7. Statutory regulations

The machine cabinet has been designed to comply with the appropriate US standards and UK legislation. However, the purchaser must take due account of any radiation protection legislation in their own countries. The following section reflects the requirements of UK legislation, which may or may not be appropriate in other countries.

Within the UK, please refer to 'Information to UK customers - Radiological Protection Information for X-ray Inspection System'. This document is available from Nordson Test and Inspection UK upon request.

1.8. Local safety rules

The user and any other personnel in the vicinity of the machine must observe all the information, instructions and procedures defined in the 'safe working arrangements' or 'local rules' in addition to the safety precautions and procedures given in this guide.

Appointed safety personnel

Safety personnel must be appointed in accordance with the safe working arrangements/local rules.

The rules may also require the appointment of an on-site Radiation Protection Supervisor (RPS) and an external Radiation Protection Advisor (RPA or Qualified Radiation Expert (QRE)).

1.9. Routine checks of X-ray cabinet

X-ray leakage from the shielded cabinet during X-ray generation must not exceed 1 μ Sv/h.

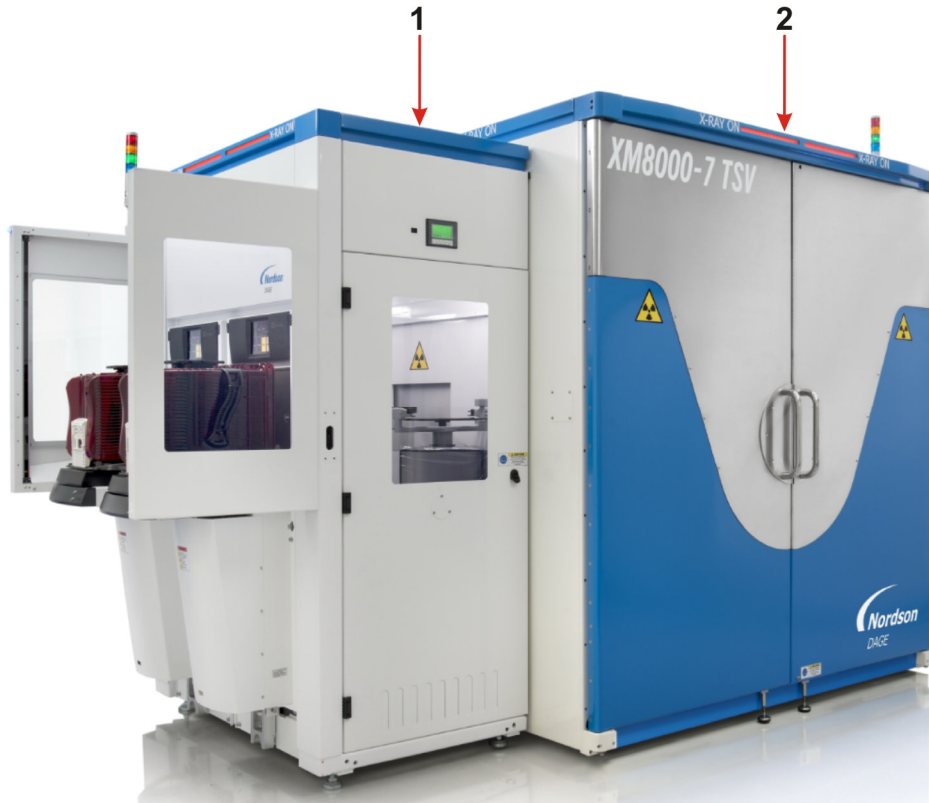
The cabinet must be checked for radiation leakage according to the procedure prepared locally by the Radiation Protection Supervisor and agreed by the Radiation Protection Advisor.

Radiation leakage should be measured using either a Geiger Muller or CW tube suitable for the measurement of 30-160kV radiation and capable of detecting radiation at levels of less than 1 μ Sv/h.

In the absence of any locally-prepared procedure, that detailed in X-ray leakage check should be performed at the stated intervals. Refer to the Maintenance Guide for details.

2. Machine overview

The XM8000 X-ray inspection machine consists of two main modules:



Part	Function
1 Brooks JCP EFEM wafer-handling system	This is a standard 2 bay / 2 load port Brooks Jet Common Platform (JCP) Equipment Front End Module (EFEM) configured to handle 300mm wafers. The Brooks enclosure incorporates a Class 1 Filter Fan Unit (FFU). This creates a positive pressure inside the Brooks enclosure. For full details on this system module, please refer to the separately-published Brooks JCP EFEM User Guide.
2 X-ray machine	The X-ray machine can be configured to carry out a range of manual, or fully-automatic inspection tests on the wafers delivered robotically from the EFEM. The machine enclosure is lead-lined and houses the X-ray imaging components.

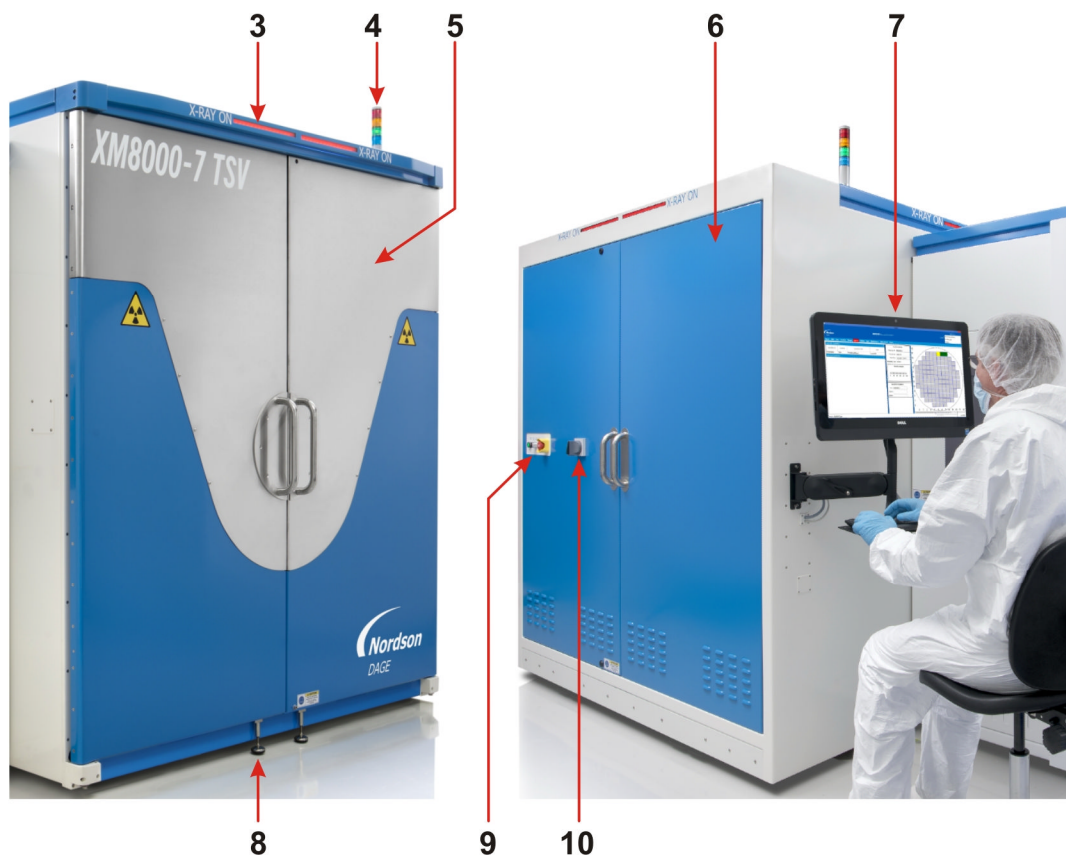
The wafers to be inspected are loaded via a Front Opening Unified/Universal Pod (FOUP).

A FOUP is a specialised plastic carrier designed to hold up to twenty-five silicon wafers securely, and to allow the wafers to be transferred safely between an Automated Material Handling System (AMHS) and the X-ray machine for processing. FOUPs typically contain a machine-readable RF tag that allows them to be identified with a unique Carrier ID.

2.1. Machine layout

The X-ray cabinet has the following main functional parts:

Note: The illustration below shows both a front and rear view of the X-ray cabinet.



Part	Function
3 X-ray warnings	The illuminated X-RAY ON warning signs on the front, side and rear walls of the machine indicate when X-rays are actually on (red).
4 Light tower	This displays the machine status using four coloured indicators, see table below for details. However, the indicators are configurable, please discuss your requirements with your local Nordson Test and Inspection representative.

Part	Function
5 Maintenance/service access doors	These doors provide full access to the interior of the lead-lined X-ray compartment. To prevent accidental opening, the doors are locked by a latch that is electrically released under software control. In the unlikely event that the door opens, the X-ray system is immediately turned off by a safety interlock system. The right-hand door is secured when closed with two front-mounted bolts.
6 Electrical and pneumatic compartment doors	The main electronic and pneumatic systems of the machine are fitted inside an accessible cabinet compartment behind hinged doors. These doors are mechanically interlocked with the main electrical isolator (10). The left-hand door is secured when closed with Allen key screws.
7 Operator workstation	This provides the means of performing and monitoring all normal operating functions. It comprises: ▪ Flat-panel monitor - connected directly to the control computer PC system, the monitor displays the operating software. ▪ Workstation tray - this supports the keyboard, and the mouse of the control computer PC system. The keyboard and mouse are used to operate all software-controlled functions of the machine. Depending on installation requirements, the operator workstation may be mounted on the EFEM cabinet.
8 Adjustable door stop	An adjustable stop is fitted on each maintenance/service access door to enable you to secure a door when open.
9 Machine control panel	This panel includes the following controls:
Power-on button	This button must be pressed in order to start the machine. It is used with the key-switch (below).

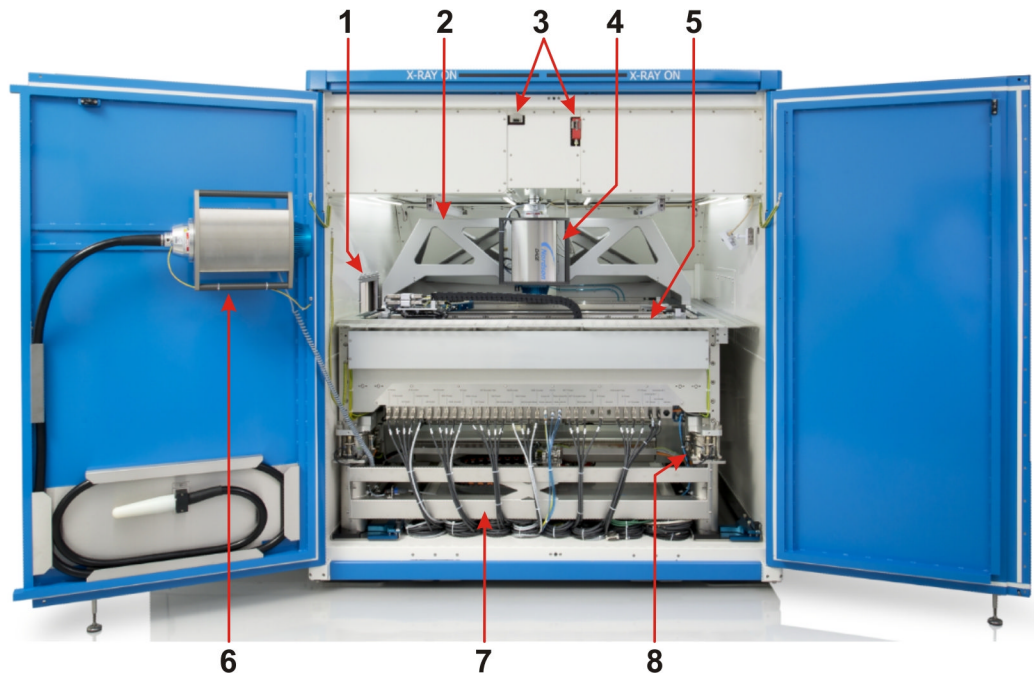
Part	Function
Key-switch	The three key-switch position functions are: Power Off - all functions of the machine are disabled. The key can be removed in this position to prevent unauthorized use. Power Enable - the control computers, pneumatics and motors are enabled, but all X-ray systems are disabled. The key can be removed in this position to prevent unauthorized use. X-Ray Enable - all functions of the machine are enabled. The key cannot be removed in this position.
Emergency Off (EMO) stop pushbutton	This emergency stop pushbutton is provided as a safety feature to be used in a hazardous situation/emergency. Operating this button instantly removes all power except the Emergency Stop pushbutton (EMO) control circuitry. Depending on machine build variant, further Emergency Off (EMO) stop pushbuttons may be mounted on the Brooks EFEM enclosure, or elsewhere on the machine cabinet. Note: To reset, rotate the red button top clockwise
10 Main electrical isolator	This lockable rotary control is used to connect or disconnect the main electrical supply to the machine.

Light tower coloured indicators

Colour	Status	Function
Red	Continuous	Currently not used.
Red	Blinking	Alarm is active.
Orange	Continuous	Load ports occupied, unavailable for transfer.
Orange	Blinking	Available for material transfer.
Green	Continuous	Idle, machine ready for use.
Green	Blinking	Job executing on machine.
Blue	Continuous	Machine is in maintenance mode.
Blue	Blinking	X-ray cabinet maintenance/service access doors open.

2.2. Interior layout

The X-ray cabinet contains the following main functional parts:

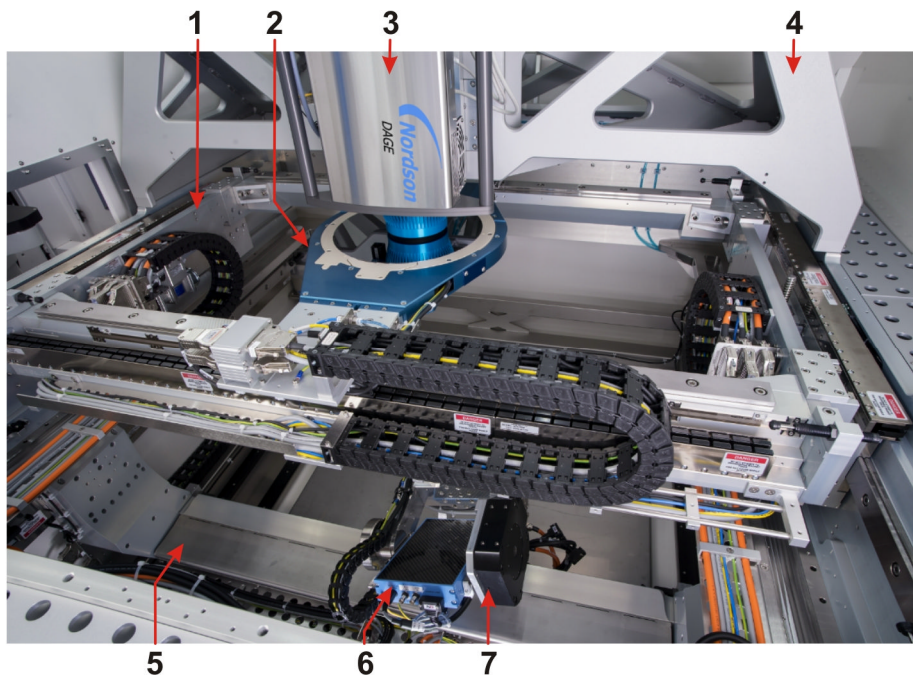


Part	Function
1 Shutter	This is a pneumatically-operated and interlocked access door between the Brooks EFEM system enclosure and the X-ray machine cabinet. When closed, it allows the user to safely use the X-ray machine while the EFEM system continues its operation cycle. When open, it allows the EFEM system to deliver a wafer to the X-ray machine wafer chuck. However, this operation can only happen if the maintenance/service access doors are closed and interlocked.
2 Upper support frame	This is a rigid aluminium frame that supports the X-ray tube and controller (4) via four anti-vibration mounts.
3 Door interlocks	There are two key-operated safety interlocks: ▪ Right-side primary - this provides an interlock with an additional solenoid that automatically locks when the maintenance/service access doors are closed. This can only be opened under software control. ▪ Left-side secondary - this provides a double-redundant interlock circuit as part of the machine safety systems.

Part	Function
4 X-ray tube and controller	The tube provides the source of X-rays and is fixed to the upper support frame (2). The tube controller is mounted on the front of the tube and contains the electronics required to control and monitor such functions as focus, voltage, and power levels.
5 Manipulator table	This assembly carries both the wafer chuck and the image detector assemblies. See below for full details on this assembly.
6 Spare X-ray tube	This is an optional spare X-ray tube.
7 Lower support frame	This is a rigid aluminium frame that supports the manipulator table (5), via four anti-vibration mounts.
8 Anti-vibration mount	These pneumatic mounts are fitted to each corner of the lower support frame and manipulator table assemblies to reduce the transmission of floor-based vibration to the X-ray imaging system components. The mounts include a height sensor/adjuster that provides for automatic self-leveling of the manipulator table under changing load conditions.

2.3. Manipulator layout

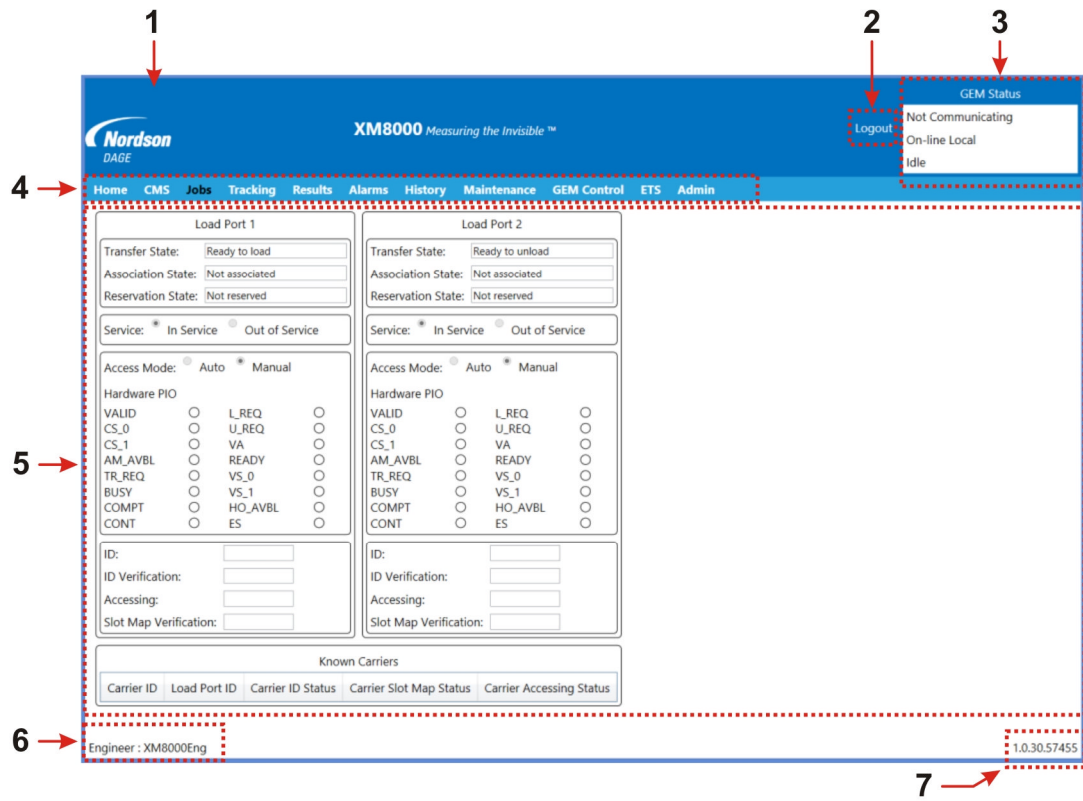
The XY manipulator mechanism contains the following main functional parts:



Part	Function
1 Wafer chuck carrier frame	This assembly carries the wafer chuck. It is mounted on precision slides and driven horizontally by linear motors in both X and Y directions. This allows different areas of the wafer sample to be accurately and repeatably positioned under the X-ray tube for examination. In addition, further rotary servo-motors are fitted to raise or lower the carrier frame Z axis.
2 Wafer chuck	This pneumatically-operated chuck receives the pre-aligned wafer delivered by the EFEM End Effector robot via the open shutter. Once delivered, the wafer is lowered under pneumatic control and clamped in place, ready for X-ray inspection.
3 X-ray tube	The tube provides the source of X-rays and is fixed to the upper support frame.
4 Upper support frame	This is a rigid aluminium frame that supports the X-ray tube and controller via four anti-vibration mounts.
5 Detector carrier frame	This assembly carries the detector carrier (7) and moves horizontally in both X and Y directions to position the image detector (6) under the wafer sample being examined. The moving assemblies of the carrier frame are mounted on precision slides and driven by linear motors.
6 Image detector	This is a CMOS high-resolution Flat Panel Detector (FPD).
7 Detector carrier	This holds the image detector (6) and is fitted to the detector carrier frame (5). Note: Depending on the system variant, the detector carrier may be optionally fitted with two motor-actuated gimbal mounts that provide both roll and pitch movement (7-axis).

2.4. Software interface

The software user-interface comprises the following functional areas:



No:	Item	Description
1	Title header bar	This identifies the application.
2	Logout control	Select this to logout the current user.
3	GEM status pane	This displays the current GEM status, such as: - Communicating - Not Communicating - On-line Local - Off-line Local - On-line Remote - Idle This pane displays a red background when the system is Not Initialized.
4	Tabbed pages bar	This displays all the available tabbed pages. Simply select the one you require and it opens in the working area. Details on the function of each tabbed page is given in the table below.

No:	Item	Description
5	Working area	This is where the software displays and operates its functional pages. The lists, controls, dialogs and other items associated with each tabbed page appear here, enabling you to carry out the necessary actions.
6	User identification	This displays the current username, and user level.
7	Software version information	This displays the software version.

Tabbed pages

Tab	This allows you to:
Home	manually initialize and stop the five main machine components (see page 25).
CMS	access the Carrier Management System page. Here you can view the status of both EFEM load ports, and any currently loaded carriers (see page 26).
Jobs	Here you can: ▪ View both process and control jobs (see page 28). ▪ Define and start a manual process (see page 28). ▪ Define and start an ad hoc process (see page 29).
Tracking	access the Substrate Tracking page. Here you can view the current status of all material in the machine (see page 31).
Results	access the inspection results. Here you can view the .csv format measurement data files (see page 31), and transfer files from the system for further analysis elsewhere (see page 35).
Alarms	access the View Active Alarms page. Here you can view both active alarms and historical alarm log data (see page 36).
History	access the View Run History page. Here you can view all runs, both active and historical process log data (see page 36).
Maintenance	access the X-ray source Maintenance and manual control pages (see page 41). Here you can manually set various parameters and perform X-ray source maintenance procedures. Note: This page is not available to an operator-level user.

Tab	This allows you to:
GEM Control	access the SECS (SEMI Equipment Communications Standard) / GEM (Generic Equipment Model) facility host to machine communication options. Here you can enable or disable facility host communication, and configure the machine to operate either: ▪ Locally under manual control, or; ▪ Remotely under facility host control.
ETS	access the Equipment Terminal Services page. Here you can send and receive text-based messages to and from the facility host system (see page 38).
Admin	access the login page. Here you can login with Operator, Engineer or Service level. It is also possible to create new users at each level and to change passwords.

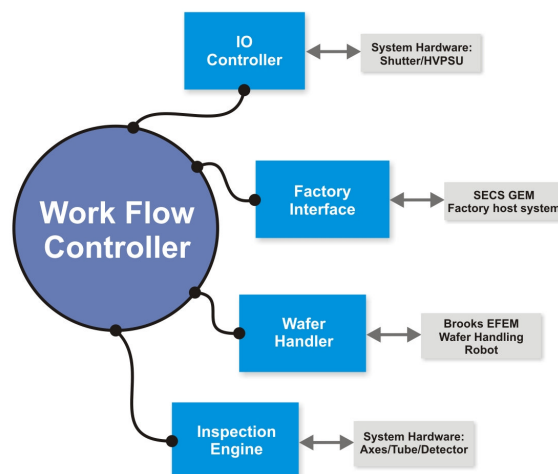
2.5. Machine components

The XM8000 machine comprises a number of function-based sub-systems. Each sub-system can include both software and hardware elements.

The following five major sub-systems have been classified as machine 'components':

Component	This represents:
INS Inspection Engine	the dedicated inspection PC and the associated mechanical, pneumatic and electrical systems that control and monitor the handling of wafers inside the XM8000 enclosure, and the acquisition of X-ray images and test results.
WFC Work Flow Controller	the dedicated workflow PC and the associated systems that control, monitor and manage the system recipes, and the handling of wafers to and from the XM8000 system. All internal communication between the various machine components is done via the WFC using Inter Process Communication (IPC). In addition, the WFC handles all process synchronisation throughout the system, and stores the test results sent from the Inspection Engine.
FI Factory Interface	the machine interface that communicates with a facility host system allowing remote control.
IO Input/Output Controller	various machine input and output interfaces and signals, such as: shutter control, contactors, safety, vacuum, HVPSU and light tower beacons.
WH Wafer Handler	the EFEM robot responsible for handling FOUPs, wafers and alignment, as well as passing the wafer through the shutter for loading on the wafer chuck.

The system components are interconnected as illustrated below:



3. System operations



WARNING

To prevent injury due to mechanical moving parts, the user should not open any of the EFEM service doors and/or the X-ray cabinet maintenance/service access doors or attempt to override safety devices and interlocks.



WARNING

To prevent death or injury caused by radiation leakage, the user should not open the X-ray cabinet maintenance/service access doors or attempt to override safety devices and interlocks.



WARNING

To prevent death or injury caused by high voltage, the user should not open the electrical and pneumatic compartment doors or attempt to override safety devices and interlocks.

During normal operation, the wafers to be inspected are loaded into a FOUP (Front Opening Unified/Universal Pod) and delivered to the XM8000 automatically, via the Automated Material Handling System (AMHS) with no user intervention required. Therefore, you would only require working access to the Equipment Front End Module (EFEM) side of the machine if you needed to manually place and remove a FOUP to and from one of the EFEM load ports.

All further functions and operations are performed either standing, or sitting at the operator workstation.

3.1. Manually loading a carrier

Wafer carriers (FOUPs) are manually loaded via one of the EFEM's two load ports.



- Load port 1 - left-hand side
- Load port 2 - right-hand side

Note: Depending on system variant, the EFEM load ports may be fitted with access doors. If this is the case, you will need to open the doors prior to loading a FOUP.

1. Place the required FOUP on an available load port.

The EFEM automatically:

- Latches and docks the FOUP.
 - Reads the integral RF tag to extract the Carrier ID for identification purposes.
 - Opens the FOUP and maps the contents.
2. Once the above process has completed, you can verify everything is correct by selecting the Tracking tab (see page 31).

3.2. Logging in

When the XM8000 X-ray machine is first started, the Home page is displayed.

1. If the Home page is not displayed, select the Home tab.
2. If no user is currently logged in, a pop up log in box is displayed.
3. Enter your Username.
4. Enter your Password.

Note: If you enter an incorrect password, you will be given a further four attempts to enter a correct password before the system locks you out. To remove a lock on your username, refer to an engineer-level user.

5. Select Log In.

6. If another user is currently logged in, do the following:

- In the Title Header Bar, select the Logout control.
- If not already selected, select the Home tab.
- Login as detailed above.

Each Username is associated with a particular level of entry defined in three main groups as follows:

- Operator - this provides the access required in order to operate the machine.
- Engineer - this provides a higher level of access to additional functionality, such as recipe creation and machine maintenance.
- Service - this level of access is limited to Nordson Test and Inspection trained field service engineers ONLY.

It is advised that the XM8000 X-ray machine is left at 'Operator' level during normal running, to allow easy access to the operational pages.

3.3. Using the Home tab

The XM8000 X-ray machine comprises a number of function-based sub-systems. The following five major sub-systems have been classified as machine 'components':

- INS - Inspection Engine
- WFC - Work Flow Controller
- FI - Factory Interface
- IO - Input/Output Controller
- WH - Wafer Handler

Each machine component can be in one of the two following states:

- **Stopped**

When a component is stopped, all its associated systems are placed in an inactive and safe state. However, they are not shut down, but are simply rendered inoperative.

- **Running**

When a component is running, all its associated systems are active and fully operational.

When the XM8000 X-ray machine is first started, all five components are stopped. Therefore, you have to manually start them running by selecting Initialize Components on the Home tab.

- If a component fails to initialize, the problem needs to be investigated before you can continue.
- Once a component has successfully initialized, the relevant entry in the on-screen component list display is updated from Stopped to Running.

If you need to stop machine operation at any time, select Stop Components. However, this action does not immediately cease machine operation or mechanical movement, as the current activity of each component is always completed, before it becomes inactive.

3.4. Viewing a load port state

This function allows you to view the operational state of each load port.

Note: There are no user controls on this page. The various displayed fields are for information only and cannot be edited.

1. Select CMS tab (Carrier Management System).

The page displays two identical, but independent Load Port panes, and a Known Carriers pane.

Each pane provides a number of information message fields and state indicators that are dynamically updated as the machine is operating. The information provided is useful when diagnosing a system fault.

Field/Indicator	Description
Load Port	
Transfer State	This shows the current transfer state: ▪ Ready to load - load port is ready to receive a carrier (FOUP). ▪ Transfer blocked - load port is currently busy. ▪ Ready to unload - load port is ready for carrier to be unloaded.
Association State	This shows the current association state: ▪ Associated - there is a carrier present on the load port. ▪ Not associated - there is no carrier present on the load port.
Reservation State	This shows the current reservation state: ▪ Not reserved - the load port is not reserved for future activity. ▪ Reserved - the load port is reserved for future activity. When in this state, the Access Mode for the port cannot be changed.
Service	These indicators show the current port service status: ▪ In service - the port is available for use. ▪ Out of service - the port is unavailable for use. This typically indicates there is currently a fault with the load port.
Access Mode	These indicators show the current port access mode: ▪ Auto - the automatic transfer of carriers to and from the load port is enabled. ▪ Manual - carriers can be manually placed and removed from the load port by a local user.

Field/Indicator	Description
Hardware PIO	These indicators show the state of the various input and output lines used by the load port and the Automatic Material Handling System (AMHS) to control carrier transfer between them.
ID	This shows the current carrier ID. This ID is either automatically read from the carrier RF tag when the carrier is loaded, or assigned as a system default if ID reading is not enabled/unavailable.
ID Verification	This shows the current ID verification state: ▪ ID Not read - the carrier RF tag has not been read. ▪ ID Verification failed - the carrier RF tag has been read, but the ID does not match the facility host system (incorrect carrier present). ▪ ID Verification OK - the facility host system has verified the carrier ID. Or alternatively, ID verification has been disabled.
Accessing	This shows the current carrier accessing state: ▪ Not accessed - the carrier has not yet been accessed by the robot. The carrier can be removed from the load port. ▪ In access - the carrier is currently being accessed by the robot. The carrier should not be removed from the load port. ▪ Carrier complete - the job involving the carrier is complete. The carrier is unlocked and ready for removal by the AMHS. ▪ Carrier stopped - the job involving the carrier is stopped. The carrier remains locked and cannot be removed by the AMHS without unlocking it first.
Slot Map Verification	This shows the current slot map verification state: ▪ Slot map not read - the carrier slot map has not yet been read. ▪ Slot map verification fail - the carrier slot map has been read, but does not match the facility host system (incorrect wafer present). ▪ Slot map verification OK - the slot map has been verified by the facility host system. ▪ Waiting for host - the XM8000 is waiting for input from the facility host. Slot map verification functionality depends on the optionally fitted OCR camera.

Known Carriers

This table displays the details of carriers already recorded in the system. Typically, it shows details of any carrier present on a load port. However, when the system is operating in Remote mode, it may show the details of carriers yet to be provided to the system in advance of their actual physical delivery.

3.5. Viewing jobs

This function allows you to view both the Process and Control jobs.

A Process Job specifies the process to be applied to a wafer. A Control Job defines what is performed on the contents of a carrier and typically comprises one or more Process Jobs.

1. Select Jobs tab > View.

The Process and Control jobs are displayed in two independent tables.

Each table may extend over more than one page, and certain table columns can be ordered and/or filtered.

2. If you wish to order the table contents, do the following:
3. Select the column heading you wish to sort by (either: Process Job ID or Control Job ID).

The results in the column will be sorted alpha-numerically and the entire table updated accordingly. The column heading shows an order direction icon:

▲ this indicates the order is ascending.

▼ this indicates the order is descending.

4. If you wish to filter the contents of a table column, do the following:

- Select the filter icon in the relevant column heading .
- Select the required filter criteria to apply: Alpha-numeric ascending, alpha-numeric descending, text filter, or text item search.
- Select OK.

3.6. Manually starting a process

This function allows you to manually execute a chosen recipe on selected wafers in a loaded carrier. To use this function:

- The system must be operating in local mode.
- The required recipe must have been previously set up.

To define and manually start a process, do the following:

1. Select Jobs tab > Manual Process Start.

The page displays two identical, but independent Load Port panes.

2. Manually load a carrier onto your chosen load port. The Carrier ID is read and displayed.

Note: The steps described below can be applied to either load port, as required.

3. Select the Recipe Type required. Choose from Inspection, Maintenance, or Image Acquisition.
4. Select the required Recipe from the drop down.
The entries shown for selection depend on the currently selected recipe type.
5. If only certain wafers in the loaded carrier are to be included in the job, do the following:
 - Set the Use selected slots radio button.
 - Set the required check boxes in the Slots list. Use the Select/Deselect All controls as necessary.
 - Enter a suitable Lot ID to identify the wafers.
 - Select Apply.
6. If only the occupied slots in the loaded carrier are to be included in the job, do the following:
 - Select the Use occupied slots radio button.
 - Enter a suitable Lot ID to identify the wafers.
7. Select Start.
The system starts automatically processing the wafers. The current system State is dynamically displayed.
8. Once the job process has finished, you can select Re-use Carrier to repeat the job using the currently loaded carrier.
Note: This control is dynamic, and is only active once the current job has finished.
9. To immediately silence the alarm buzzer, select Stop Buzzer.

3.7. Managing an ad hoc process

This function allows you to choose which wafers in the loaded carrier are to be included in the job, and the particular recipe to be used for an individual wafer. Each wafer is then processed a preselected number of times. To use this function:

- The system must be operating in local mode.
- The required recipes must have been previously set up.

To define and manually execute an ad hoc job, do the following:

1. Select Jobs tab > Ad Hoc.
The page provides two identical, but independent Load Port tabs on the left-hand side, and a process report list on the right-hand side.
2. Manually load a carrier on to your chosen load port. The Carrier ID is read and displayed.
Note The steps described below can be applied to either load port, as required.
3. Select the load port tab you require.

4. If you want to include one or more individually configured wafers in the job, do the following in the slot list:
 - Set the check box against a required wafer entry.
 - Enter the wafer Lot ID.
 - Select the Recipe Type required.
Choose from Inspection, Maintenance, or Image Acquisition.
 - Select the required Recipe from the drop down.
The entries shown for selection depend on the currently selected recipe type.
 - Select the number of Iterations for the wafer.
 - If you want to be able to choose a new recipe after the first recipe run is completed, set the Manual Override check box.
Once the first recipe run is completed (all iterations), a message is displayed prompting you to select a new recipe, or return the wafer to the carrier.
5. If you want to apply a specific configuration to all the wafers, do the following:
 - In the slot list, set the check box against a wafer entry. Typically, this would be entry 1 by default, but can be any entry.
 - Enter the wafer Lot ID.
 - Select the Recipe Type required.
Choose from Inspection, Maintenance, or Image Acquisition.
 - Select the required Recipe from the drop down.
The entries shown for selection depend on the currently selected recipe type.
 - Select the number of Iterations for the wafer.
 - Select the 'as Slot' from the drop down. This must match the slot entry you have just configured.
 - Select CopyAll.
6. Select the number of Iterations for the job.
7. Select Start.
8. To stop the ad hoc wafer processing job at any point, click Stop.
Stopping the process is a controlled recovery procedure that immediately turns the X-rays off, and returns the wafer to the carrier. This function might be required if an incorrect recipe has been selected and started.
9. To abort the ad hoc wafer processing job at any point, click Abort.
Aborting the process immediately turns the X-rays off, but does not return the wafer to the carrier. This function might be required if a critical (but not emergency) situation arose during a job. However, it is recommended that in most situations Stop is used.

10. Once the job process has finished, you can select Re-use Carrier to repeat the job using the currently loaded carrier.

Note: This control is dynamic, and is only active once the current job has finished.

11. To immediately silence the alarm buzzer, select Stop Buzzer.

3.8. Tracking a substrate

This function allows you to track the substrates (wafers) currently loaded in either load port.

Note: Apart from the list selection tabs, there are no user controls on this page. The various displayed fields are for information only and cannot be edited.

1. Select Tracking tab.

The page provides two identical, but independent tabular Load Port Substrate lists on the left-hand side. Each list can be shown or hidden, as required. However, you can only show one list at a time.

A Current Substrate report is displayed on the right-hand side, it shows a number of substrate parameters, job statistics and equipment status information.

2. Select the required Load Port Substrates tab.

The tabular list shows the Location, State and Jobs for each substrate (wafer).

The substrate can be in one of four Locations:

- A numbered carrier slot.
- Tool - this is the X-ray machine wafer chuck.
- Robot - this is the EFEM End Effector robotic wafer handler.
- Aligner - this is the EFEM aligner mechanism.

3.9. Viewing job results

The XM8000 X-ray machine stores the results of each inspection run in an ordered series of folders that are named as follows:

Recipe id > Carrier id_[Run id] > Slot number.

For each slot number a further series of folders are created to hold the actual results, as follows:

Folder name	Description
BumpLevel	This stores Comma Separated Value (.CSV) text files that contain information about bumps with defects only. Information about the files is found here: BumpLevel folder files (see page 33).

Folder name	Description
DieLevel	This stores CSV text files that contain information about every single bump on that die - both with, or without defects. The number of files corresponds to the number of inspection locations. Information about the files is found here: DieLevel folder files (see page 33).
Images	This stores the processed images taken from the selected locations in JPEG format.
The above folders are created for all inspection runs. However, the following two folders are only created for the more comprehensive recipes:	
WaferLevel	This stores CSV text files that contain additional information about the inspection run. Information about the files is found here: WaferLevel folder files (see page 34).
WaferMaps	This stores CSV text files that contain information about the position of the scanned dies within the wafer.

In addition, at the slot number folder level, the system stores the following files, when required:

Filename	Description
[Recipe id]_DIEMap	This JPEG image file shows the position of dies on the wafer.
VoidReport1_[Run id]	This CSV text file lists the total number of voids found during the inspection run throughout the whole wafer, and puts them into two categories. Information about the file format is found here: VoidReport1 example (see page 35).
VoidReport2_[Run id]	This CSV text file lists all detected voids. Each void is listed as a new line entry. Information about the file format is found here: VoidReport2 example (see page 35).

This function allows you to navigate to, and view the stored results.

1. Select Results tab > Results.

The page displays the following three interrelated panes:

- A Currently Available Results table - by default, this lists all the job results in Run id order.
- A navigation tree on the left-hand side - here you can navigate through the various result folders. You can expand or collapse each folder level, as required.
- A folder contents window on the right-hand side -

The contents of these panes are dynamically updated as selections are made.

2. If you wish to filter the contents of the results table, do the following:
 - Select the filter icon in the relevant column heading .
 - Select the required filter criteria to apply.
3. Select the required result entry in the Currently Available Results table.
4. Navigate through the folder levels displayed in the left-hand navigation tree.
5. Select the required results file once it is displayed in the right-hand window.

BumpLevel folder file format

Column	Description
BumpID	This is a unique bump identifier within a die.
VoidID	This is a unique identifier for void.
Bump_X & Bump_y	Position of a bump in Wafer coordinate system. It is exactly the same values as for Center_X Center_Y in WBA_Die_Results.
X (µm), Y (µm)	Void location presented in Axis Coordinate System to help the user navigate to the exact location of the bump using analyser dialog if necessary.
Diameter_X & Diameter_Y	Horizontal and Vertical diameter of a void.
Diameter %	$\text{DiameterOfVoid} / \text{DiameterOfBump} * 100\%$. DiameterOfVoid - average void diameter, DiameterOfBump-average bump diameter.
Area (µm²)	Void area in µm².
Area (%)	$\text{VoidArea} / \text{BumpArea} * 100\%$. VoidArea is total void area (sum of all voids in a bump)

DieLevel folder file format

Column	Description
BumpID	This is the unique bump identifier in a die.
Flag	Takes values: ▪ AS_UNKNOWN = -1 ▪ AS_ABSENT = 0 ▪ AS_OK = 1 ▪ AS_MISSING = 2 ▪ AS_BRIDGED = 3
Failure	(-1) if the bump is processed and 0 otherwise. The bump can have 0 only for Comprehensive recipe (has CAD), in case bump specified in cad cannot be found. For Basic recipe it is always 1.

Column	Description
Diameter	Diameter of a bump. It is the average diameter of a bump.
Area	Area of bump in μm .
Roundness	Roundness of Bump. A bump that is a perfect circle would have roundness of 100.
Void Fraction	Voiding Area / Bump Area, the value are always below 1. VoidingArea is total voiding area (sum of areas of all voids in bump).
Largest Void Fraction	Similar to Void Fraction but taking into account only one (the largest void).
VoidNumber	This is the number of voids in a bump.
Center_X and Center_Y	This is the position of a bump in Wafer Coordinate System.
LargestVoidDiameterFraction	DiameterOfLargestVoid/DiameterOfBump.Value up to 1. Only the largest void is taken into account.

WaferLevel folder files

The folder can contain one or more of the following CSV files:

Filename	Description
WBA_DieMap	This lists the dies and their coordinates in Wafer Coordinate System (Centre_x and Centre_Y)
In the following, Centre_X and Centre_Y are the die coordinates from WBA_DieMap	
WBA_LargestVoidArea	This lists the maximum void area in each die.
WBA_LargestVoidDiameterPercent	This lists the maximum void diameter in each die, expressed as a percentage.
WBA_LargestVoidPercent	This lists the maximum largest void in each die, expressed as a percentage.
WBA_Roundness_Min	This lists the minimum roundness of all bumps in a die.
WBA_VoidNumberTotal	This lists a total count of voids in each die.
WBA_Diameter_Avg	This lists the average diameter of all bumps from all dies on the wafer (only included if flag is OK).
WBA_Diameter_Max	This lists the maximum diameter of all bumps in a die.
WBA_Diameter_Min	This lists the minimum diameter of all bumps in a die.
WBA_Diameter_Stdev	This lists the standard deviation of the diameters of all bumps in a die.

VoidReport1 Format

The void report contains the following columns:

- Recipe
- Lot
- Wafer
- Void Percentage
- Void Count (by area)
- Void Count (by diameter)

A row entry might look like (for example only):

v30Rel_Hybrid_Unq_AREA,20180806_1,ORANGEF1.25,10.0~23.4%,0,0

VoidReport2 Format

The void report contains the following columns:

- Recipe
- Lot
- Wafer
- MAX Area Void
- MAX Diameter Void
- Total NG
- Dield
- BumpId
- Area Percent
- Diameter Percent

3.10. Transferring job results

This function allows you to configure the XM8000 X-ray machine to transfer results data files to a specified location either automatically, or manually.

To configure the target destination, do the following:

1. Select Results tab > Data File Transfer.
2. To change the Current Network Transfer Path from the displayed value, do the following:
 - Enter the required New Network Path.
 - Select Update.

To manually transfer a results file to the target destination, do the following:

1. Select Results tab > Data File Transfer.
2. Ensure the network transfer path has been configured.
3. Select the Lot ID from the drop down.
4. Select Transfer.

3.11. Viewing system alarms

This function allows you to view the XM8000 X-ray machine alarms.

1. Select Alarms tab.
2. Select Active Alarms to display any currently active alarms.
3. Select Alarm Log to display historical alarms.
4. Select the time period you require:
 - Last Hour
 - Last Day
 - Last Week
 - All
5. Select Apply.

The alarms are displayed in a chronologically-ordered table.

The table may extend over more than one page, and the table columns can be filtered.

6. If you wish to filter the contents of a table column, do the following:
 - Select the filter icon in the relevant column heading .
 - Select the required filter criteria to apply.
7. To immediately silence the alarm buzzer at any time, select Stop Buzzer.

3.12. Viewing job history

This function allows you to view the job history.

1. Select History tab.
2. Select the time period you require:
 - Day
 - Week
 - All
3. Select Apply.

The jobs are displayed in a table.

The table may extend over more than one page, and the table columns can be filtered.

4. If you wish to filter the contents of a table column, do the following:
 - Select the filter icon in the relevant column heading .
 - Select the required filter criteria to apply.

3.13. Configuring local or facility host control

This function allows you to configure the XM8000 X-ray machine control states.

1. Select GEM Control tab.
2. Select one of the following Control options, as required:
 - Local - manual control only.
 - Remote - facility host system control is also allowed.
3. Select On-line or Off-line, as required.

The GEM Status pane is updated to reflect the current setting:

- On-line Local - all manual functions are available to the user.
- Off-line Local - limited manual functions are available to the user.
- On-line Remote - all manual functions are available to the user. In addition, the facility host system can remotely operate the machine through the communications interface.

3.14. Configuring communications

This function allows you to configure the communication state between the XM8000 X-ray machine and the facility host system.

1. Select GEM Control tab.
2. Select one of the following Communications options, as required:
 - Enable - the machine can communicate with the facility host system using the SECS-II (SEMI Equipment Communications Standard, Part 2) communication protocol.

Enable has two sub-states: Communicating - this indicates that communications have been established and are active. Not Communicating - this indicates that communications have not yet been established.

- Disable - the machine cannot communicate with the facility host system.

When you switch from Enable to Disable, all communication ceases immediately, any unsent messages, actions and open transactions are either discarded, or terminated.

The GEM Status pane is updated to reflect the current setting:

- Communicating - Communications are enabled and established.
- Not Communicating - Communications are either disabled, or enabled but not established.

3.15. Managing host and user messages

This function allows a user to acknowledge messages sent from the facility host system, and send messages to the facility host system.

When a message is sent to the XM8000 X-ray machine, the ETS (Equipment Terminal Services) tab becomes active and flashes red to alert the current user that an incoming message requires attention, and acknowledgment.

All messages sent via this mechanism are logged by the facility host system. The XM8000 X-ray machine is not involved in any form of interpretation of the message data, and does not process the message data in any way.

1. Select ETS tab.

The Host and Operator Comms are displayed in two independent panes.

2. When a message is received from the facility host and displayed in the Host Comms pane, do the following:
 - Note (and where necessary, act on) the content of the message.
 - Select Acknowledge to send a time-stamped acknowledgment receipt to the facility host system that confirms an XM8000 user has viewed the message, and de-activate the flashing tab.
3. If you need to send a message to the facility host, do the following:
 - Type the message in the Operator Comms pane.
 - Select Send.

3.16. Changing a password

This function allows an engineer-level user to change any user's password. If you are logged in as an operator-level user, you cannot access this function. Therefore, if you need to change your password, refer to an engineer.

1. Select Admin tab > Change Password.

By default, the current user's username is automatically selected and displayed.

2. If you need to change another user's password, select his or her username from the drop down.
3. Enter the existing Password.
4. Enter the New Password.

It is recommended that passwords are alpha-numeric as standard practise, and include a mixture of upper and lowercase characters. Password length is not checked.

5. Enter the new password exactly as entered above, into the Confirm New Password field.
6. Select Change Password.

3.17. Managing users

This function allows an engineer-level user to add new users, and unlock any locked users. If you are logged in as an operator-level user, you cannot access this function.

To add a new user, do the following:

1. Select Admin > Manage Users.
2. Enter the new Username.
3. Select an appropriate Role for the user:
 - Operator - this provides the access required in order to operate the machine.
 - Engineer - this provides a higher level of access to additional functionality, such as machine maintenance.
 - Service - this level of access is limited to Nordson Test and Inspection trained field service engineers ONLY.
4. Enter the New Password for the user.

It is recommended that passwords are alpha-numeric as standard practise, and include a mixture of upper and lowercase characters. Password length is not checked.

5. Enter the new password exactly as entered above, into the Confirm New Password field.
6. Select Add User.

The new user is added to the user table.

To unlock a locked user, do the following:

1. Select Admin > Manage Users.
2. Locate, and select the relevant locked user in the user table.
3. Right-click on the selected user, and select Unlock.

3.18. Specifying log capture

This function allows you to acquire all the XM8000 X-ray machine diagnostic files, and any selected Windows™ event logs for a chosen date. This information can then be used when troubleshooting a system fault.

1. Select Admin tab > About.
2. Select the date and time of the incident you are interested in.
3. Set the check box against any required Windows Event Log type.
4. Select Capture Logs.

The steps in the capture logs process are displayed in a scrollable window. Once completed, a dated incident zip file that contains all the selected log files is created in the following folder:

C:\Users\Public\Documents\Dage\XM\IncidentLogs\

4. System maintenance operations

4.1. Maintenance control tab

The maintenance tab allows an engineer-level user to enter and exit the maintenance mode. Simply select the Enter or Exit control, as required.

In addition, this tab also provides access to the maintenance-based functions listed in the table below.

Note: Access to the maintenance tab is only allowed when a job is not running, and the X-rays are off.

Function	This provides access to:
Equipment Status	the following dynamically-updated X-ray parameters: ▪ Voltage (kV) ▪ Power (W) ▪ Uptime (hours)
Manual Control	dynamically-updated machine control status information, and the controls necessary for allowing the manual operation of various system components, such as the EFEM: ▪ load ports ▪ robot ▪ aligner ▪ shutter
Recovery	the controls that allow you to stop and initialize the five major sub-systems classified as machine 'components'.
Door Control	the controls that allow you to open and close the X-ray enclosure maintenance/service access doors.
IO Control	a range of function-specific machine indicators and manual controls for the: ▪ Shutter ▪ Wafer Chuck ▪ Door
Load Port Usage	manual controls and state indicators for both load ports.
Equipment Constants	a number of machine-dependant variables that can be altered to support different operational situations.

4.1.1. Equipment status

The equipment status page provides dynamically-updated machine information, as detailed below:

- X-Ray Voltage (kV)
- X-Ray Power (W)
- X-Ray Uptime (hours)

4.1.2. Manual control

The manual control page is divided into five discrete sections. Each section is dedicated to a particular machine functional area and provides both dynamically-updated status information, and the controls necessary for allowing manual operation of the various system components and robotics, as detailed in the tables below.

Note: Each section has its own message pane that displays any applicable system messages.

When used together, the controls provided allow you to carry out a range of tasks such as:

- Clamping, loading and mapping a carrier.
- Moving an individual wafer from a source location to a chosen destination location.
- Opening and closing the EFEM to X-ray cabinet shutter, and loading or unloading the wafer chuck.

1 - Load Port 1 + 2 Control section

Display/Control	Description
Re-use Carrier	Select the appropriate control to repeat the maintenance task using the currently loaded carrier.
Clamp/Unclamp Carrier	Select the appropriate control to clamp or unclamp the carrier (FOUP) currently loaded. The current clamp status is displayed: ▪ Unclamped ▪ Clamped
Load/Unload Carrier	Select the appropriate control to load or unload the carrier (FOUP) currently loaded. The carrier is docked, opened and mapped ready for use. The current load status is displayed: ▪ NO CARRIER ▪ Loaded

Display/Control	Description
Map Status	This displays the current carrier map status: ▪ Unknown ▪ Mapped

2 - Robot Control section

Display/Control	Description
Source:	Select the source location of the wafer you wish to use. The prefixes LP1 and LP2 are used to identify specific load ports. The list includes only system locations where a wafer is actually present, such as: ▪ Mapped load port carrier slot ▪ Aligner ▪ Inspection (X-ray machine wafer chuck) The list also includes the substrate (wafer) ID.
Destination:	Select the destination location of the wafer you have selected. The list includes only system locations where no wafers are present, such as: ▪ Empty load port carrier slot ▪ Aligner ▪ Inspection (X-ray machine wafer chuck) Notes: ▪ It is recommended that wafers are ALWAYS pre-aligned prior to being passed to the X-ray machine. ▪ If the shutter is closed, the Inspection location is not available.
Move	Select to move the wafer from the source to the destination location. Once a move operation has completed, the source, destination and occupancy display fields on this page are dynamically updated.

3 - PreAligner section

Display/Control	Description
Occupancy:	This displays the current occupancy status of the aligner: ▪ EMPTY ▪ Current substrate (wafer) ID
Value field	Select the alignment angle value

Display/Control	Description
AlignWafer	Select to align the wafer. The current alignment status is displayed: - Unknown - Aligned

4 - Inspection Control section

Display/Control	Description
Occupancy:	This displays the current occupancy status of the X-ray machine: - EMPTY - Current substrate (wafer) ID
Load/Unload	Select to load/unload the wafer chuck.
Inspection	Moves the wafer chuck to the inspection position (directly under the X-ray tube).

5 - Shutter Control section

The shutter can only be operated manually when the machine's safety system requirements have been fully met.

Display/Control	Description
Open Shutter	Select to open the EFEM to X-ray cabinet shutter.
Close Shutter	Select to close the EFEM to X-ray cabinet shutter.

4.1.3. Recovery

The recovery page allows you to stop and initialize the five major sub-systems classified as machine 'components', using the manual controls detailed in the table below.

Note: The initialize and stop functions provided here are the same functions as provided on the Home tab (see page 25).

Control	Description
System Recovered	Select to attempt to recover the system from a failure state. However, in most cases it is recommended that you stop the components and re-initialize when a failure occurs (see below).
Initialize Components	Select to initialize the five machine components. If a component fails to initialize, the problem needs to be investigated before you can continue. Once a component has successfully initialized, the relevant entry in the on-screen component list display is updated from Stopped to Running.

Control	Description
Stop Components	Select to stop the five machine components. This function terminates all machine operation. However, this action does not immediately cease machine operation or mechanical movement, as the current activity of each component is first completed, before it becoming inactive.
Shutdown PCs	Select to shutdown the three system PCs (WFC, IO, INS).

4.1.4. Door control

The door control page provides the manual controls that allow you to open (unlock) and close (lock) the X-ray enclosure maintenance/service access doors.

Control	Description
Open Door	Select to unlock the maintenance/service access doors. This function simply unlocks the door interlock. The two bolts fitted to the right-hand door will still need to be removed, and the doors physically opened. All X-ray generation is terminated, and the machine put into a non-recoverable state that requires a complete restart.
Close Door	Select to lock the maintenance/service access doors. This function simply locks the door interlock. The doors will still need to be physically closed and the right-hand door secured by the front-mounted bolts.

4.1.5. IO Control

The IO Control page provides a number of function-specific machine IO state indicators, and the manual controls detailed in the tables below.

Note: The shutter and door can only be operated manually when the machine's safety system requirements have been fully met.

The IO indicators allow you to see any changes in state dynamically during the manual exercising of the shutter, wafer chuck and door locks, as well as providing useful diagnostic machine state information in the event of a failure.

Shutter and wafer chuck

Control	Description
Open	Select to open the shutter, or wafer chuck.
Close	Select to close the shutter, or wafer chuck.
Cycle	Select to start a cycle of open and close operations.
Stop	Select to stop any currently running cycle operation.

Door

Control	Description
Open Door	Select to unlock the maintenance/service access doors. This function simply unlocks the door interlock. The two bolts fitted to the right-hand door will still need to be removed, and the doors physically opened.
Close Door	Select to lock the maintenance/service access doors. This function simply locks the door interlock. The doors will still need to be physically closed and the right-hand door secured by the front-mounted bolts.

4.1.6. LOAD port usage

The load port usage page provides the following manual controls and indicators for both load ports:

Control/Indicators	Description
In Service	Select to set the load port as available for use.
Out of Service	Select to set the load port as unavailable for use. You would typically use this to indicate if there is a fault with the load port.
Access Mode: Auto	Select to enable the automatic transfer of the carriers (FOUPs) to and from the load port via an Automatic Material Handling System (AMHS).
Access Mode: Manual	Select to enable carriers to be manually placed and removed from the load port by a local user.
Continue Manual Carrier Delivery	Select to allow a carrier that has been manually placed on a load port currently in Auto access mode, and with a transfer state Ready To Load - to be processed. Normally, when this situation arises, the system raises an access violation alarm and treats the transfer as incomplete. However, this function allows you to proceed with manual operation.
PIO Reset	Select to clear any PIO alarms related to the load port.
Hardware PIO	This shows the state of the various inputs and output lines used by the load port and the AMHS to control carrier transfer between them.

4.1.7. Equipment constants

If enabled, the host can read and update a number of equipment constant. This page provides information on, and allows the enabling of these specific equipment constants parameters.

Note: These parameters must not be changed unless under the strict guidance of Nordson Test and Inspection.

Control	Description
Drop down	This lists the equipment constants, select as required. Set the value to: ▪ 0 = Disabled ▪ 1 = Enabled
Update	Select to save the change.

5. Troubleshooting

Every customer facility will have its own methods for problem recovery. However, when the machine is in operation it is envisaged that the user will not be responsible for any maintenance, servicing, or repair. Therefore, if an operational fault develops, always consult an authorised supervisor, engineering level user, or a member of your facility's technical/maintenance staff to resolve the issue.

Typical faults might be as follows:

- System fails to power ON
- System powers OFF
- System reports vacuum error
- System reports low air pressure
- Wafer slipping from wafer handling robotics
- Timeout alarms for wafer handling robotics
- Any errors generated by the system during normal operation

6. Index

A

About this guide • 6
Appointed safety personnel • 10

C

Changing a password • 38
Configuring communications • 37
Configuring local or facility host control
• 37
Contact information • 4

D

Document signal words • 3
Door control • 45

E

Equipment constants • 47
Equipment status • 42

F

Foreword • 3

H

Hazardous Energies • 7

I

Interior layout • 15
IO Control • 45
Ionizing radiation • 9

L

Labels • 4
LOAD port usage • 46
Local safety rules • 10
Logging in • 24

M

Machine components • 21
Machine layout • 12
Machine overview • 11
Maintenance control tab • 19, 41

Managing an ad hoc process • 19, 29
Managing host and user messages • 20,
38

Managing users • 39

Manipulator layout • 16

Manual control • 42

Manually loading a carrier • 24

Manually starting a process • 19, 28

O

Operators safety rules • 7

R

Recovery • 44
Routine checks of X-ray cabinet • 10

S

Safety information • 3
Safety interlocks • 8
Shielding methods • 9
Software interface • 18
Software license • 5
Specifying log capture • 39
Statutory regulations • 10
System maintenance operations • 41
System operations • 23

T

Tracking a substrate • 19, 24, 31
Transferring job results • 19, 35
Troubleshooting • 49

U

Using the Home tab • 19, 25, 44

V

Viewing a load port state • 19, 26
Viewing job history • 19, 36
Viewing job results • 19, 31
Viewing jobs • 19, 28
Viewing system alarms • 19, 36

X

X-ray On indicators • 9

X-ray safety • 9